

Master IASD – Intelligence Artificielle, Systèmes, Données

Data Science Lab – Project A2

VAE Pre-training and Kernel Herding for a Fixed-Capacity GAN

Effect on quality, diversity and training cost on MNIST

Group 24

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Papers studied

Unbalanced GANs: Pre-training the Generator of Generative Adversarial Network using Variational Autoencoder

H. Ham, T. J. Jun, D. Kim — arXiv 2020

Super-Samples from Kernel Herding

Y. Chen, M. Welling, A. Smola — UAI 2010

Abstract

We investigate VAE pre-training and kernel herding to improve MNIST generation under a fixed generator architecture. While VAE pre-training alone offers limited benefits, kernel herding substantially improves the GAN baseline, reducing FID from 24.36 to 12.41 and increasing precision. Jointly selecting samples from the classical GAN and the VAE-GAN further reduces FID to 8.13 and increases recall from 0.375 to 0.474, at a modest precision cost.

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1 Introduction

The A2 constraint fixes the generator architecture, an MLP of about 1.49M parameters, for MNIST generation assessed by Fréchet Inception Distance (FID), precision and recall. We investigate two complementary levers: changing the generator's initialisation and training through variational autoencoder (VAE) pretraining [1], and correcting its output distribution at inference through kernel herding [2]. We first assess these interventions separately against a long-run GAN baseline, then ask whether a *joint selection* from the conventionally trained GAN and a VAE-initialised GAN can exploit their complementary samples. Our focus is the quality–diversity trade-off rather than architectural capacity.

2 Experimental framework

Setup. MNIST images are scaled to $[-1, 1]$. The supplied generator is unchanged: $100 \to 256 \to 512 \to 1024 \to 784$ with LeakyReLU and tanh; the discriminator is unconstrained. Unless otherwise stated, we use the starter's binary cross-entropy (BCE) adversarial losses, Adam ($\text{lr} = 2 \cdot 10^{-4}$, $\beta_1 = 0.9$), batch size 64 and 100 epochs. The original GAN/VAE training comparisons use seed 0, paired initialisation and fixed latent codes. New short fine-tuning and mixed-pool experiments are separately identified below; their results must not be interpreted as independent training-seed replications.

Metrics. Each reported full evaluation uses 10,000 samples and the starter's `generate.py` conversion (negative values clipped to zero, then 8-bit quantisation); replacing this by rescaling to $[0, 1]$ worsens the original baseline FID from 27.3 to 35.1. FID [3] uses 20,000 fixed MNIST training samples. Precision and recall use the k-NN support estimator [4] with $k = 3$ and 10,000 real images; they measure geometric support, not classification accuracy or class proportions. We also report class entropy from a small CNN (test accuracy 98.7%) as a diagnostic. The original local and platform evaluations agree approximately on FID (27.3 versus 28.1) but differ in precision/recall (0.47/0.34 versus 0.52/0.23); comparisons below are therefore made within our fixed local protocol.

3 Baseline and training dynamics

We run the starter GAN for 300 epochs (seed 0), evaluated every 50 epochs. Its epoch-100 checkpoint is the reference for the VAE factorial study; epoch 150 is used for the subsequent inference-time selections.

Observations. FID falls from 313 at initialisation to 27.3 at epoch 100, reaches its minimum at epoch 150 (24.5; precision 0.55, recall 0.36), and deteriorates to 73.5 at epoch 300. Precision peaks at 0.60 (epoch 250) before falling to 0.42, whereas recall remains around 0.34–0.38. Class entropy decreases from 0.96 to 0.53 as classifier confidence increases from 0.94 to 0.98. At epoch 100, digit 1 accounts for 19.3% of generated samples (11.2% of training data), but digit 2 for only 2.6% (9.9% in training data), as shown in Figure 1.

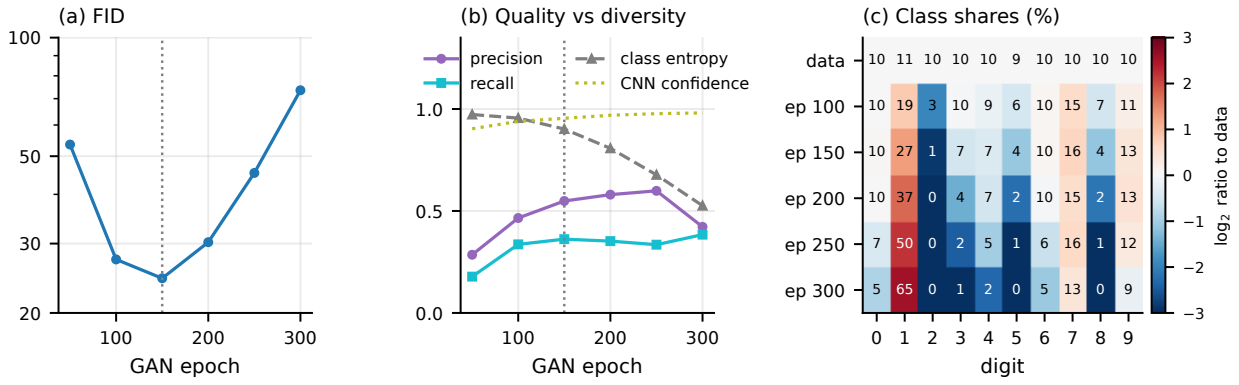


Figure 1: Long GAN baseline (seed 0): (a) FID, (b) precision, recall, class entropy and confidence, (c) digit shares over training. The epoch-150 model is the subsequent herding baseline.

Interpretation (hypothesis). The discriminator loss falls from 1.01 to 0.52 between epochs 100 and 300; rising confidence and falling entropy suggest that the generator concentrates on easier classes. Support-based k-NN recall need not reflect this frequency imbalance, while FID depends on distributional moments. We did not test this mechanism directly. Longer training does not resolve the trade-off, motivating inference-time selection (Section 5).

4 VAE pretraining of the generator

Method. Following Ham et al. [1], we train a VAE [5] whose decoder is *exactly* the provided generator, then copy its weights into G and initialise a fresh discriminator and optimiser. The encoder uses widths $784 \to 1024 \to 512 \to 256$, dropout 0.25, and predicts the mean and log-variance of $q_\psi(z \mid x)$; the prior is $p(z) = \mathcal{N}(0, I_{100})$. With decoder G_θ, the objective is

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{x \sim P_{\text{MNIST}}} \left[\mathbb{E}_{z \sim q_\psi(\cdot|x)} \text{BCE}\left(\frac{G_\theta(z)+1}{2}, \frac{x+1}{2} \right) + D_{\text{KL}}(q_\psi(z \mid x) \,\|\, \mathcal{N}(0, I)) \right]. \tag{1}$$

We minimise it for 50,000 steps (about 53 epochs; Adam, learning rate $2 \cdot 10^{-4}$, $\beta_1 = 0.5$, batch 64). The decoder transfer is checked to be exact. We cross VAE initialisation with discriminator regularisation (D-reg: dropout 0.25 in D and $\beta_1 = 0.5$ for both optimisers). D-reg changes two factors together, so its effect cannot be attributed to either one separately. Pretraining adds 53 epochs to the compute budget.

Table 1: Original 2×2 study, seed 0, at GAN epoch 100. E_{40}: GAN epochs until FID stays below 40; total with VAE pretraining in parentheses.

VAE init.	D-reg	FID ↓	P ↑	R ↑	E_{40}
No	No	27.3	0.47	0.34	63
No	Yes	26.0	0.48	0.36	49
Yes	No	37.7	0.41	0.26	91 (145)
Yes	Yes	30.3	0.42	0.32	85 (139)

Starting point. The pretrained decoder alone reaches FID 28.2, precision 0.37 and recall 0.34, close to the epoch-100 baseline in FID and recall but worse in precision. Its effective latent dimension is only 5.6 (99.4 at random initialisation), estimated by

$$d_{\text{eff}} = \frac{\left(\sum_i s_i\right)^2}{\sum_i s_i^2}, \qquad s_i = \mathbb{E}_z \left\| \frac{\partial G_\theta(z)}{\partial z_i} \right\|_2^2. \tag{2}$$

Results. The pretrained generator leads early (epoch 10: FID 142 versus 213 with standard D, 109 versus 122 with D-reg) but is worse than its control from epoch 20 onward (Figure 2). Standard adversarial training is unstable: D loss is around 0.05 at epochs 10–30, recall is zero at epoch 10, and entropy falls to 0.07 by epoch 20. D-reg avoids this collapse (entropy above 0.98 after initialisation), yet FID still rises from 28 to 109 during the first ten epochs. At epoch 100, VAE initialisation costs +10.4 FID without D-reg and +4.3 with D-reg. Including pretraining, reaching FID 40 takes 2.2–2.3 times as many epochs as the corresponding baseline. D-reg alone accelerates convergence (49 versus 63 epochs), and helps the VAE by preventing severe early collapse.

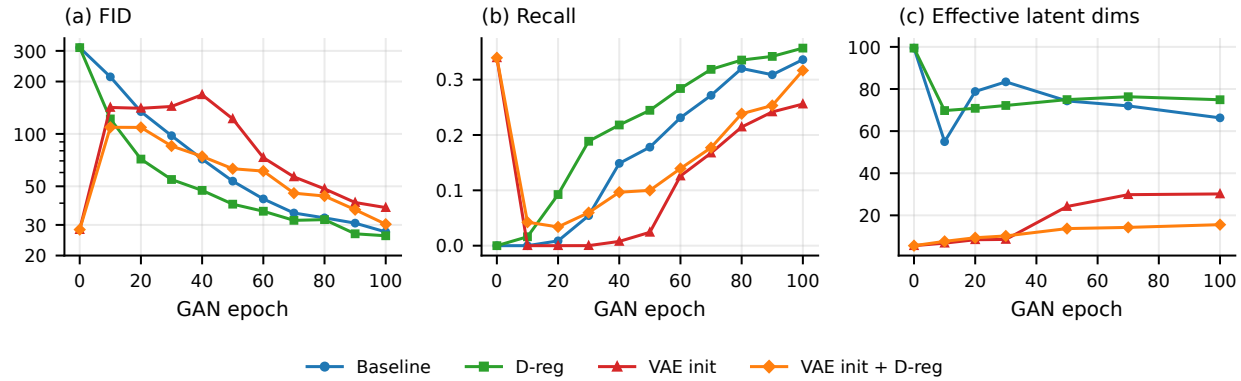


Figure 2: Four conditions in Table 1, evaluated every ten GAN epochs: (a) FID, (b) recall, (c) effective latent dimensions.

Why is the VAE start lost? A fresh discriminator distinguishes frozen VAE-decoder samples from real digits more easily than trained-baseline samples (AUC 0.97 versus 0.87 after 100 steps), suggesting that blur is an accessible cue. D-reg reduces the generator gradient norm from 6.8 to 0.25, but outputs on fixed latents still move markedly (0.78 versus 0.41 RMS). Lowering lr_G to $2 \cdot 10^{-5}$ slows the initial deterioration (FID 36.5 rather than 109 at epoch 10), but the run ends at FID 43.4, below neither the unmodified fine-tuning nor the decoder's initial score. Latent usage alone does not account for recall:

at epoch 100, effective dimensions are roughly 30 (VAE) and 16 (VAE+D-reg) versus 66–75 without VAE, while the corresponding VAE recalls are 0.26 and 0.32.

Take-away. Pretraining gives a strong standalone decoder, but under the *long adversarial schedules tested here* it does not improve the final GAN. This motivates a distinct, short reconstruction-preserving fine-tuning strategy in Section 6, rather than claiming that VAE-initialised generators can never help.

5 Kernel herding at inference

Method and distributional interpretation. Let $\phi(x) \in \mathbb{R}^{2048}$ be the Inception-v3 embedding, P_ϕ the distribution of real MNIST embeddings and Q_ϕ the distribution of selected generated embeddings. With the Gaussian radial-basis-function kernel

$$k_\sigma(u, v) = \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right), \qquad \sigma = 14.286, \tag{3}$$

we compare distributions using the squared maximum mean discrepancy (MMD):

$$\mathrm{MMD}_k^2(P_\phi, Q_\phi) = \mathbb{E}_{u,u' \sim P_\phi} k(u, u') - 2\mathbb{E}_{u \sim P_\phi, v \sim Q_\phi} k(u, v) + \mathbb{E}_{v,v' \sim Q_\phi} k(v, v'). \tag{4}$$

The Gaussian RBF kernel is *characteristic*: at population level, $\mathrm{MMD}_k(P_\phi, Q_\phi) = 0$ if and only if $P_\phi = Q_\phi$. Thus minimising MMD aims to match the *MNIST distribution in Inception feature space*; it does not guarantee equality of pixel distributions, because ϕ need not be injective.

For $M = 2048$ real reference features $r_i = \phi(x_i)$ and T selected features $s_j = \phi(\tilde{x}_j)$, Monte Carlo / empirical plug-in estimation of (4) gives

$$\widehat{\mathrm{MMD}}_k^2 = \frac{1}{M^2} \sum_{i,i'=1}^{M} k(r_i, r_{i'}) - \frac{2}{MT} \sum_{i=1}^{M} \sum_{j=1}^{T} k(r_i, s_j) + \frac{1}{T^2} \sum_{j,j'=1}^{T} k(s_j, s_{j'}). \tag{5}$$

The first term is independent of the selected set. For the equal-weight empirical measure, greedily adding one candidate to reduce (5) is equivalent, for the RBF kernel ($k(x, x) = 1$), to choosing

$$\tilde{x}_{t+1} \in \arg\max_{x \in \mathcal{C} \setminus S_t} \left\{ \underbrace{\frac{1}{M} \sum_{i=1}^{M} k(\phi(x), r_i)}_{\text{attraction to MNIST}} - \underbrace{\frac{1}{t+1} \sum_{j=1}^{t} k(\phi(x), \phi(\tilde{x}_j))}_{\text{repulsion from earlier picks}} \right\}. \tag{6}$$

This is the greedy kernel-herding criterion [2]: its target mean embedding is estimated by Monte Carlo over MNIST training features. We use a median-distance bandwidth, no repeated indices, and select $T = 10{,}000$ images from 20,000 GAN candidates. Finite references and a restricted candidate pool mean that the characteristic-kernel property provides an objective, not a guarantee of recovering the true distribution.

Table 2: Epoch-150 GAN: selecting 10,000 from one pool of 20,000 candidates. Random: mean ± standard deviation over five selections; time: selection only. The submitted KH row is a distinct original run.

Selection	FID ↓	P ↑	R ↑	Entropy	Time (s)
Random	24.2 ± 0.4	0.549 ± 0.002	0.369 ± 0.006	0.903	–
Attraction only	17.7	0.646	0.257	0.935	8
Herding, Eq. (6)	12.3	0.613	0.372	0.949	167
Stochastic herding (1000/step)	12.3	0.612	0.372	0.949	183
Submitted selection	12.4	0.612	0.375	0.947	–

Ablation and mechanism. Herding roughly halves FID and raises precision while maintaining recall (Table 2). Attraction alone acts as a quality filter: it raises precision to 0.646 but lowers recall to 0.257. The repulsion term restores coverage and improves class balance (entropy 0.903 to 0.949), although digit 2 remains underrepresented (roughly 1.4% to 1.9%, versus 9.9% in MNIST). No selector can produce a missing mode if the candidate pool lacks it. Stochastic herding matches full herding in this experiment but saves little time because the implementation still updates repulsion against all candidates.

6 Joint herding of complementary generators

Short VAE-GAN fine-tuning. We independently reproduced the 50,000-step VAE pretraining recipe and transferred its decoder weights into the same prescribed generator architecture. To retain reconstruction information during GAN fine-tuning, we froze the pretrained encoder and minimised for the generator

$$\mathcal{L}_G = \underbrace{\mathrm{BCE}(D(G(z)), 1)}_{\mathcal{L}_{\mathrm{adv}}} + \lambda_{\mathrm{rec}} \underbrace{\mathrm{BCE}\left(\frac{G(z_q)+1}{2}, x_{01}\right)}_{\mathcal{L}_{\mathrm{rec}}}, \quad z_q \sim q_\psi(\cdot \mid x), \tag{7}$$

where $z \sim \mathcal{N}(0, I_{100})$, $x_{01} \in [0,1]^{784}$ and q_ψ encodes the corresponding MNIST image scaled to $[-1,1]$. We use $\lambda_{\mathrm{rec}} = 2.5$, 50 *optimizer steps* (not epochs), batch 128, Adam $\mathrm{lr}_G = \mathrm{lr}_D = 5 \cdot 10^{-5}$, $\beta_1 = 0.5$, and a fresh discriminator. Unconditional sampling for evaluation still uses $z \sim \mathcal{N}(0, I_{100})$. This differs from the 100-epoch fine-tuning study above and is not directly a compute-matched comparison.

Mixed pools. We combine samples from the epoch-150 GAN and the short-fine-tuned VAE-GAN. Both have identical architectures but *different weights*; the joint method therefore uses two checkpoints. For each GAN/VAE fraction (75/25, 65/35, 50/50), we create one fixed pool of 20,000 candidates (disjoint latent indices across sources), then apply a *single* Inception-RBF herding selection of 10,000 images with Eq. (6). The same $M = 2048$ real training features, $\sigma = 14.286$, output quantisation and local Inception-2048 FID/P/R protocol are retained. The results in Table 3 come from existing fixed candidate pools, not independent training repetitions.

Table 3: Joint GAN/VAE-GAN selection, each from a pool of 20,000 candidates to 10,000 final images. $k = 3$; FID and precision/recall from the same local evaluator.

Candidate pool	Selection	FID ↓	P ↑	R ↑
GAN only	Random	24.362	0.549	0.371
GAN only	KH	12.414	**0.612**	0.375
VAE-GAN only	Random	24.961	0.398	0.377
VAE-GAN only	KH	12.435	0.507	0.331
GAN/VAE 50/50	Random	18.477	0.472	**0.518**
GAN/VAE 75/25	Joint KH	8.637	0.590	0.479
GAN/VAE 65/35	Joint KH	8.305	0.588	0.476
GAN/VAE 50/50	Joint KH	**8.132**	0.582	0.474

Results. Despite inferior single-generator precision, the VAE-GAN contributes complementary samples. Relative to GAN-only KH, the 50/50 joint KH lowers FID by 34.5% (12.414 to 8.132) and raises recall from 0.375 to 0.474, while precision decreases from 0.612 to 0.582. The 75/25 mixture has the highest precision (0.590) and recall (0.479) *among the joint-KH mixes*, at FID 8.637; the small differences across ratios should not be overinterpreted. Joint KH retains 5,029 GAN and 4,971 VAE-GAN images for the 50/50 pool. Crucially, random selection from that *same* pool has higher recall (0.518) but markedly worse FID (18.477) and precision (0.472): mixing provides candidate diversity; herding improves the quality–coverage compromise of the selected output, not recall in isolation.

Qualitative inspection. Figure 3 compares samples from the individual generators and their herded selections. The GAN generally produces sharper digits, while the VAE-GAN shows more ambiguous strokes. Joint herding retains samples from both models.

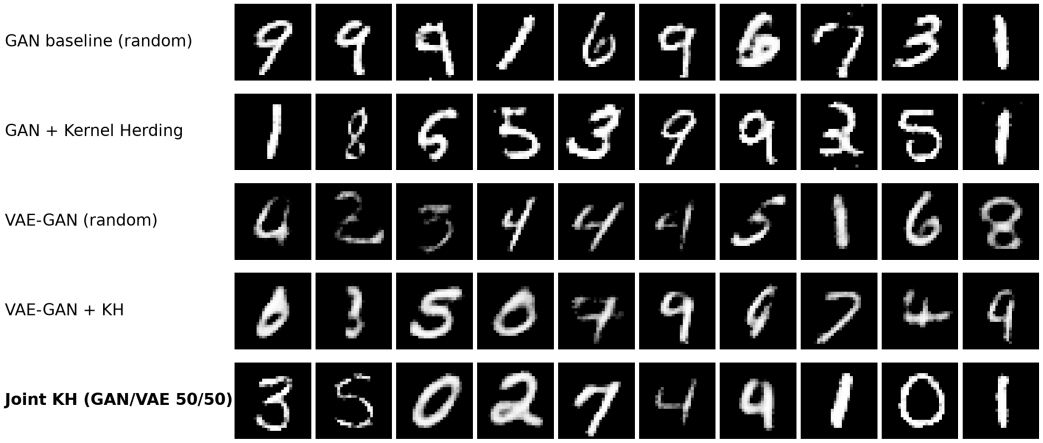


Figure 3: Qualitative comparison of MNIST generations: GAN, GAN+KH, VAE-GAN, VAE-GAN+KH, and joint GAN/VAE kernel herding (50/50). Ten random examples per configuration.

7 Discussion and limitations

Two levers, complementary effects. Under the longer training schedules, VAE pretraining fails to improve the GAN baseline; inference-time herding on the epoch-150 GAN roughly halves FID while preserving its recall. Joint herding, by contrast, benefits from two different candidate distributions. The improvement cannot be attributed to VAE alone or to herding alone: both the mixed random control and the separate-generator herding controls are necessary to isolate their effects.

Scope and cost. Because herding optimises a kernel discrepancy in Inception space and FID is evaluated in that same embedding, some of the gain may be feature-specific. In the original GAN ablation, using real training references disjoint from those used for FID gave nearly unchanged FID (12.3 versus 12.4); this does not rule out feature-space bias. Herding requires twice as many candidates as final images and $O(|\mathcal{C}|T)$ kernel-score updates in our implementation. The mixed method also uses two sets of generator weights, increasing *total inference capacity* even though each network has the required architecture. Finally, the original five random draws quantify selection variability for a fixed generator, not training variability; the new mixtures share fixed caches and have not been revalidated across independent pools or platform evaluation. The reported values describe local experimental results, not guaranteed reproducible platform scores.

8 Conclusion

Within the prescribed MLP architecture, kernel herding was the strongest single-generator intervention: it improved epoch-150 GAN FID from 24.36 to 12.41 and precision from 0.549 to 0.612 with little change in recall. VAE pretraining alone was not consistently beneficial under prolonged adversarial fine-tuning. Yet a short reconstruction-regularised VAE-GAN provided complementary candidates: joint herding over a 50/50 pool achieved our lowest measured FID (8.13), with recall 0.474 versus 0.375 for GAN-only herding, at a modest precision cost. These findings show that training and inference-time selection can complement each other, while highlighting the importance of the comparison protocol, feature-space choice and total inference capacity.

References

[1] H. Ham, T. J. Jun and D. Kim. *Unbalanced GANs: Pre-training the Generator of Generative Adversarial Network using Variational Autoencoder*. arXiv:2002.02112, 2020.

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[5] D. P. Kingma and M. Welling. *Auto-Encoding Variational Bayes*. ICLR, 2014.